

Katherin Liced Molina Granados

Computational Researcher — Autonomous Systems, Multi-Agent Systems, Computational Modelling

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Research Profile

Computational researcher with an interdisciplinary background in Information Technology, Business Analytics, and software engineering. Current research focuses on coordination, communication, and distributed decision-making in heterogeneous autonomous systems operating under uncertainty.

Develops computational frameworks combining Multi-Agent Systems, Agent-Based Modelling, distributed artificial intelligence, experimental design, and software engineering — examining how autonomous software agents, robotic platforms, and human supervisors share mission information, allocate tasks, coordinate resources, and adapt when operational conditions change.

Long-term research objective: generalizable frameworks for real-time mission communication, adaptive task execution, and collaborative autonomy in complex field environments.

Education

Mas- ter	Pontificia Universidad Católica de Chile , Business Analytics	Chile
	- Graduated with Maximum Distinction; GPA 6.7/7.0 - Ranked 1st in Graduation Seminars I, II, and III; overall ranking 4th of 40 graduates - Research on the integration of machine learning, topological segmentation, survival analysis, explainable AI, and financial simulation	Jan 2024 – Mar 2026
Diplom	Pontificia Universidad Católica de Chile , Business Intelligence	Chile
	GPA 6.6/7.0 — business intelligence, data analytics, decision-support systems, data warehousing, organisational performance analysis	Jan 2024 – Dec 2024
Bach- elor of Infor- ma- tion Tech- nol- ogy	Victoria University , Web & Mobile Application Development	Australia
	- GPA 6.13/7.0 — software development, systems analysis, databases, object-oriented programming, computer networks - Capstone: designed, engineered, and deployed a full social network platform for Victoria University, end to end	Jan 2020 – Dec 2023
Un- der- grad- uate (5 se- mes- ters)	Universidad Cooperativa de Colombia , Telecommunications and Electronics Engineering	Colombia
	Foundational studies in electronics, telecommunications, mathematics, and engineering systems	Jan 2009 – Dec 2011

Experience

Kahlo Group Pty Ltd, CPTO

Leads technology, analysis, and research-oriented initiatives involving organisational systems, software architecture, business processes, and applied artificial intelligence.

- Development of the AOOS (Autonomous Organization Operating System) research and technology programme
- Design of architectures for autonomous software agents and investigation of multi-agent coordination and governance mechanisms
- Development of structured workflows, information models, and decision systems; integration of research, engineering, and product-development practices

Sydney, Australia
Jan 2015 – present
11 years 9 months

KITBC Consulting, Consultant Data Analyst

Short-duration analytics engagement (client Intelligent Chile) supporting operational improvement for large-scale automated mining operations.

- Analyzed operational information to identify performance bottlenecks and process inefficiencies
- Direct exposure to autonomous fleet operations, operational analytics, and the interaction between autonomous technology and operational management

Chile
Feb 2025 – Feb 2025
1 month

Research and Technology Development

AOOS — Autonomous Organization Operating System

Kahlo Group. Industrial research and technology development programme combining investigation of autonomous cognitive and multi-agent systems with the engineering of an enterprise software platform for autonomous organisations.

- Multi-agent orchestration, autonomous cognitive systems, and agent communication protocols
- Agent capability/role representation and organisational knowledge ontologies
- Auditability, decision traceability, explainable agent behaviour, and human-in-the-loop supervision

Jan 2023 – present

Selected Publications

Structural Coordination Capacity, Not Incentive Intensity, Governs Decentralized Heterogeneous Multi-Robot Missions: Evidence from the MOSAIC Agent-Based Model

Katherin Molina, Lorena Holguin

zenodo.org/records/21718799 (Zenodo preprint)

July 2026

MOSAIC v1.1.0 — Mission-Oriented Self-Organization through Auctions, Incentives, and Coalitions

Katherin Molina

www.comses.net/codebases/2174fbc6-55cd-4f6e-abc7-9a79aa25ef81/releases/1.1.0 (CoMSES Computational Model Library (peer-reviewed))

Aug 2026

A Multi-Layer Analytical Framework for Customer Retention in Voluntary Health Insurance

Katherin Molina

zenodo.org/records/19815868 (Zenodo preprint)

Apr 2026

Research Skills

Computational Research: Problem formulation, experimental design, Agent-Based Modelling, computational simulation, statistical analysis, model verification and validation, reproducible research

Autonomous & Multi-Agent Systems: Mission-level coordination, distributed decision-making, agent communication, adaptive task allocation, coalition formation, market-based coordination, human-autonomy teaming

AI & Analytics: Machine learning, explainable AI, topological data analysis, behavioural segmentation, survival analysis, business analytics

Programming & Software Engineering: Python, R, SQL, NetLogo, Git/GitHub, software architecture, data pipelines

Analytical Methods: XGBoost, Random Survival Forest, Kaplan–Meier, Cox proportional hazards, SHAP, UMAP, HDBSCAN, financial simulation

Robotics Foundations: Localisation, path planning, search algorithms, autonomous navigation, ROS concepts

Professional Memberships

Australian Computer Society — Associate Member, No. 4448962

IEEE (Institute of Electrical and Electronics Engineers) — Member, No. 101391129

Research Preparation

Introduction to Agent-Based Modeling — Santa Fe Institute, Complexity Explorer (2026)

Robotics Software Engineer Nanodegree — Udacity

Languages

Spanish: Native

English: Full Professional Proficiency